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FINITE-COEFFICIENT GERSTEN INJECTIVITY FAILS IN RAMIFIED MIXED CHARACTERISTIC

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Abstract. Let V be a complete discrete valuation ring of mixed characteristic $(0,3)$ in which 3 is a uniformizer, and put $A = V[\![x,y]\!]/(3+x^2-y^3)$. We construct a nonzero class $a \in K_2(A;\mathbf{Z}/3)$ whose restriction to the fraction field of A is zero. Thus Gersten injectivity for algebraic K-theory with $\mathbf{Z}/3$-coefficients fails for a two-dimensional ramified regular local ring. We also indicate, without proof, the expected analogous statement in degree $p-1$ for every odd prime p. This counterexample does not contradict the integral Gersten conjecture but it rules out a naive reduction to finite coefficients.

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1. Introduction

For a ring R, write $K(R)$ for nonconnective algebraic K-theory and $K(R;\mathbf{Z}/3) := K(R)\wedge \mathbf{S}/3$, where $\mathbf{S}/3$ is the Moore spectrum. For every odd prime p, we have $p\cdot \mathrm{id}_{\mathbf{S}/p} = 0$ in the stable homotopy category. Hence the groups $K_n(R;\mathbf{Z}/p)$ are naturally $\mathbf{F}_p$-modules; in particular, multiplication by $1/2$ on a $\mathbf{Z}/3$-coefficient group means multiplication by 2. We have $K_n(R;\mathbf{Z}/3) = \pi_n K(R;\mathbf{Z}/3)$, and the coefficient sequence contains

$$K_2(R) \xrightarrow{3} K_2(R) \longrightarrow K_2(R;\mathbf{Z}/3) \xrightarrow{\partial_R} K_1(R) \xrightarrow{3} K_1(R). \tag{1}$$

Here finite coefficients mean the spectrum-level Moore construction $K(R)\wedge\mathbf{S}/3$, applied term by term to the Gersten row. This should not be confused with derived tensoring a fixed integral Gersten complex with $\mathbf{Z}/3$.

Our main result is the following explicit counterexample.

Theorem 1.1. *Let V be a complete discrete valuation ring of mixed characteristic $(0,3)$ in which 3 is a uniformizer, and let k be its residue field*[1]*. Set*

$$A := V[\![x,y]\!]/(3+x^2-y^3), \qquad F := \mathrm{Frac}(A). \tag{2}$$

Then A is a complete two-dimensional ramified regular local domain and there is a class

$$0 \neq a \in \ker\big(K_2(A;\mathbf{Z}/3) \longrightarrow K_2(F;\mathbf{Z}/3)\big).$$

Moreover $K_1(A)[3] = K_1(F)[3] = 0$, so this is also a failure of injectivity of $K_2(A)/3 \to K_2(F)/3$.

Consequently the augmented finite-coefficient Gersten row is not exact at its first term. By contrast, Gersten exactness with arbitrary finite coefficients holds for unramified regular local rings [Ska17, Cor. 5.7]. This does not contradict the equicharacteristic integral results of Quillen and Panin, Gillet's finite-coefficient theorem for

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[1] In other words, V is a Cohen ring [The26, Tag 0327]; for instance $V = \mathbf{Z}_3$, or $V = W(k)$ for a perfect field k of characteristic 3.

discrete valuation rings, or the smooth mixed-characteristic theorem of Gillet-Levine [Qui73, Pan03, Gil86, GL87].

The ring in (2) is ramified: its maximal ideal is (x, y) and $3 = y^3 - x^2 \in (x, y)^2$. Its special fibre is the cusp $A/3A \simeq k[\![x, y]\!]/(x^2 - y^3)$, so A is regular but $V \to A$ is not ind-smooth. Recent positive results for smooth or ind-smooth algebras over valuation rings likewise do not contain this example [BK25, Kun24].

There is one literature consequence worth making precise. After introducing an arbitrary cohomology theory with supports, [Lüd24, Conj. 1.1] asserts that its Cousin complex resolves the associated Zariski sheaf on every regular scheme. In that literal unrestricted form the conjecture is false: take the theory with supports $H^q_Z(X) := K^Z_{-q}(X; \mathbf{Z}/3)$ and apply Theorem 1.1 to the local scheme $\mathrm{Spec}(A)$. The actual theorem of that paper concerns p-adic étale Tate twists on schemes smooth over a henselian discrete valuation ring and is unaffected [Lüd24, Thm. 1.2]. Similarly, the explicitly "imprecise" broad formulation in [Kun24, Conj. 1.1] cannot be read as including mod-p algebraic K-theory in residue characteristic p on all regular rings; the proved integral statements there remain unchanged.

The proof is short once two inputs are isolated. The class itself is elementary: the prime divisor $y = 0$ of $\mathrm{Spec}(A)$ has coordinate ring $A/yA \simeq O := V[\zeta]$, and a is the pushforward along $\mathrm{Spec}(O) \hookrightarrow \mathrm{Spec}(A)$ of a mod-3 class of O. Its generic restriction vanishes because it is supported on a divisor. Nonvanishing is proved after the flat base change $A \to B[1/3]$, where B is the normal surface $uv = y^3$ obtained from A by adjoining ζ. After this base change, the divisor splits into two conjugate points and the class becomes the product of a mod-3 class with a line bundle of order three in $\mathrm{Pic}(B[1/3])$. This product is detected in weight two of the motivic filtration. The motivic argument identifies the obstruction and controls its passage from the associated graded to K-theory. It also isolates the higher-degree pattern behind the proposed extension in Remark 5.2.

2. Motivic detection of Bott products with line-bundle classes

We use the following consequence of the multiplicative motivic filtration on algebraic K-theory. In the equicharacteristic case relevant below, the filtration and its motivic graded pieces were constructed by Elmanto-Morrow [EM23, Thm. 1.1]; Bouis extended this construction beyond equicharacteristic [Bou24, Thm. C]. The filtered first Chern class and the ind-smooth comparison used below are supplied by [Bou25, BK25].

Lemma 2.1. *Let T be a connected regular affine noetherian scheme of dimension one, ind-smooth over a field of characteristic zero. Set $F^q := \mathrm{Fil}^q_{\mathrm{mot}} K(T)$ and $F^q_3 := \mathrm{Fil}^q_{\mathrm{mot}} K(T; \mathbf{Z}/3) = \mathrm{cofib}(3\colon F^q \to F^q)$. Denote by $i_q\colon F^q \to F^0 \simeq K(T)$ and $i_{q,3}\colon F^q_3 \to F^0_3 \simeq K(T; \mathbf{Z}/3)$ the structure maps of the two decreasing filtrations, and by $e_q\colon F^q \to \mathrm{gr}^q_{\mathrm{mot}} K(T)$ and $e_{q,3}\colon F^q_3 \to \mathrm{gr}^q_{\mathrm{mot}} K(T; \mathbf{Z}/3)$ the quotient maps. Then the following assertions hold.*

(i) *The cofiber sequences defining the spectra F^q_3 assemble into a cofiber sequence of filtered spectra*

$$\mathrm{Fil}^\bullet_{\mathrm{mot}} K(T) \xrightarrow{3} \mathrm{Fil}^\bullet_{\mathrm{mot}} K(T) \longrightarrow \mathrm{Fil}^\bullet_{\mathrm{mot}} K(T; \mathbf{Z}/3) \xrightarrow{\widetilde{\partial}} \Sigma\, \mathrm{Fil}^\bullet_{\mathrm{mot}} K(T). \tag{3}$$

For every $q \geq 0$, applying gr^q to (3) gives the motivic coefficient triangle

$$\mathbf{Z}(q)_{\mathrm{mot}}(T)[2q] \xrightarrow{3} \mathbf{Z}(q)_{\mathrm{mot}}(T)[2q] \to \mathbf{Z}/3(q)_{\mathrm{mot}}(T)[2q] \xrightarrow{\delta_q[2q]} \mathbf{Z}(q)_{\mathrm{mot}}(T)[2q+1].$$

Thus, on homotopy groups, $(e_q)_* \widetilde{\partial}_* = (\delta_q[2q])_* (e_{q,3})_*$.

(ii) *The map $(i_{1,3})_*\colon \pi_2 F^1_3 \to K_2(T; \mathbf{Z}/3)$ is an isomorphism. If $\zeta \in \mu_3(T)$ and $\beta \in K_2(T; \mathbf{Z}/3)$ satisfy $\partial_T(\beta) = [\zeta]$, define $\widetilde{\beta} := (i_{1,3})_*^{-1}(\beta)$. Under the canonical identification $\pi_2 \mathrm{gr}^1_{\mathrm{mot}} K(T; \mathbf{Z}/3) = H^0_{\mathrm{mot}}(T, \mathbf{Z}/3(1)) = \mu_3(T)$, one has $(e_{1,3})_*(\widetilde{\beta}) = \zeta$.*

(iii) *The homomorphism $(i_1)_*\colon \pi_0 F^1 \to K_0(T)$ is injective with image $\ker(\mathrm{rk}\colon K_0(T) \to \mathbf{Z})$. For a line bundle $\mathcal{L}$ on T, let $\widetilde{\lambda} \in \pi_0 F^1$ be the unique class satisfying $(i_1)_*(\widetilde{\lambda}) = [\mathcal{L}] - 1$. Then $(e_1)_*(\widetilde{\lambda}) = c_1^{\mathrm{mot}}(\mathcal{L})$ in $\pi_0 \mathrm{gr}^1_{\mathrm{mot}} K(T) = H^2_{\mathrm{mot}}(T, \mathbf{Z}(1))$.*

(iv) *The motivic-to-étale comparison maps $\mathrm{comp}_q\colon \mathbf{Z}/3(q)_{\mathrm{mot}}(T) \to R\Gamma_{\text{Ét}}(T, \mu_3^{\otimes q})$ are multiplicative. Moreover, $H^0(\mathrm{comp}_1)(\zeta) = \zeta$ and $H^2(\mathrm{comp}_1)(c_1^{\mathrm{mot}}(\mathcal{L}) \bmod 3) = \kappa(\mathcal{L})$, where $\kappa(\mathcal{L}) \in H^2_{\text{Ét}}(T, \mu_3)$ is the image of $[\mathcal{L}]$ under the boundary map of the Kummer sequence.*

Proof. By [Bou24, Defs. 3.18 and 3.20, Prop. 4.46], the spectra F^q form a functorial multiplicative decreasing $\mathbf{N}$-indexed filtration of $K(T)$, with $F^0 \simeq K(T)$ and $\mathrm{gr}^q_{\mathrm{mot}} K(T) \simeq \mathbf{Z}(q)_{\mathrm{mot}}(T)[2q]$. Evaluation at a fixed

filtration degree and the functor gr^q are exact functors on filtered spectra. Hence the levelwise cofibers F_3^q form the cofiber sequence (3), and

$$\mathrm{gr}^q_{\mathrm{mot}} K(T;\mathbf{Z}/3) \simeq \mathrm{cofib}\big(3\colon \mathrm{gr}^q_{\mathrm{mot}} K(T) \to \mathrm{gr}^q_{\mathrm{mot}} K(T)\big) \simeq \mathbf{Z}/3(q)_{\mathrm{mot}}(T)[2q]. \tag{4}$$

Exactness also identifies $\mathrm{gr}^q(\widetilde{\partial})$ with $\delta_q[2q]$, including its commutative square with the quotient maps. The multiplication gives maps $F_3^r \wedge F^s \to F_3^{r+s}$. Under (4), their associated-graded maps are

$$\mathbf{Z}/3(r)_{\mathrm{mot}}(T)[2r] \otimes^{\mathbf{L}} \mathbf{Z}(s)_{\mathrm{mot}}(T)[2s] \longrightarrow \mathbf{Z}/3(r+s)_{\mathrm{mot}}(T)[2r+2s],$$

induced by the multiplication of the motivic complexes. This proves (i).

Choose a field k_0 of characteristic zero over which T is ind-smooth. The classical-motivic comparison is an equivalence on T by [BK25, Thm. 6.1]. Thus $\mathbf{Z}(0)_{\mathrm{mot}}(T) \simeq R\Gamma_{\mathrm{Zar}}(T,\mathbf{Z})$ and $\mathbf{Z}(1)_{\mathrm{mot}}(T) \simeq R\Gamma_{\mathrm{Zar}}(T,\mathbf{G}_m)[-1]$; see also [Bou24, Examples 3.4 and 3.5].

We record the weight-zero consequence in the degrees that will be used. Since T is connected and regular, it is integral. The constant Zariski sheaves $\mathbf{Z}$ and $\mathbf{Z}/3$ on T are consequently flasque, and their cohomology is concentrated in degree zero. Consequently $\mathbf{Z}(0)_{\mathrm{mot}}(T) \simeq \mathbf{Z}$, and $\mathbf{Z}/3(0)_{\mathrm{mot}}(T) \simeq \mathbf{Z}/3$ concentrated in degree zero. It follows from (4) that, for $j > 0$, $\pi_j \mathrm{gr}^0_{\mathrm{mot}} K(T;\mathbf{Z}/3) = 0$. Under the classical-motivic comparison, the map $\pi_0 F^0 = K_0(T) \to \pi_0 \mathrm{gr}^0_{\mathrm{mot}} K(T) = \mathbf{Z}$ is the usual rank homomorphism. The fibre sequence $F_3^1 \to F_3^0 \to \mathrm{gr}^0_{\mathrm{mot}} K(T;\mathbf{Z}/3)$ gives the exact segment

$$0 = \pi_3 \mathrm{gr}^0_{\mathrm{mot}} K(T;\mathbf{Z}/3) \to \pi_2 F_3^1 \xrightarrow{(i_{1,3})_*} K_2(T;\mathbf{Z}/3) \to \pi_2 \mathrm{gr}^0_{\mathrm{mot}} K(T;\mathbf{Z}/3) = 0.$$

Thus $(i_{1,3})_*$ is an isomorphism.

The integral weight-zero fibre sequence similarly gives $(i_1)_*\colon \pi_1 F^1 \xrightarrow{\sim} K_1(T)$, because $\pi_2 \mathrm{gr}^0_{\mathrm{mot}} K(T) = \pi_1 \mathrm{gr}^0_{\mathrm{mot}} K(T) = 0$. Define the weight-one edge homomorphism $\epsilon_1\colon K_1(T) \to H^1_{\mathrm{mot}}(T,\mathbf{Z}(1))$ as $(e_1)_* \circ (i_1)_*^{-1}$. The compatibility between the filtered orientation and the motivic first Chern class, used in [Bou25, Rem. 4.4] and made explicit in [EM23, Lemma 5.16], identifies the restriction of ϵ_1 along $\Gamma(T,\mathcal{O}_T)^\times \to K_1(T)$ with the map on H^1 induced by

$$c_1^{\mathrm{mot}}\colon R\Gamma_{\mathrm{Nis}}(T,\mathbf{G}_m)[-1] \longrightarrow \mathbf{Z}(1)_{\mathrm{mot}}(T)$$

of [Bou25, Def. 4.1 and Rem. 4.2] (see also [EM23, Lemma 5.1 and Definition 5.2]). Under the classical-motivic comparison on T, this map on H^1 is the identity of $\Gamma(T,\mathcal{O}_T)^\times$. In particular, $\epsilon_1([\zeta]) = \zeta$.

Naturality of the boundary in (3) gives $(i_1)_*\widetilde{\partial}_*(\widetilde{\beta}) = \partial_T(\beta) = [\zeta]$. Therefore $\widetilde{\partial}_*(\widetilde{\beta})$ is the unique element of $\pi_1 F^1$ mapped to $[\zeta]$ by $(i_1)_*$.

Put $\bar{\beta} := (e_{1,3})_*(\widetilde{\beta})$. Part (i) and the preceding equality give $\delta_1(\bar{\beta}) = \zeta$ in $H^1_{\mathrm{mot}}(T,\mathbf{Z}(1)) = \Gamma(T,\mathcal{O}_T)^\times$. Under $\mathbf{Z}(1)_{\mathrm{mot}}(T) \simeq R\Gamma_{\mathrm{Zar}}(T,\mathbf{G}_m)[-1]$, the long exact sequence of the weight-one coefficient triangle identifies $H^0_{\mathrm{mot}}(T,\mathbf{Z}/3(1))$ with $\mu_3(T)$, and identifies δ_1 with the inclusion $\mu_3(T) \hookrightarrow \Gamma(T,\mathcal{O}_T)^\times$. Consequently $\bar{\beta} = \zeta$, proving (ii).

The integral weight-zero fibre sequence also gives the exact sequence $0 = \pi_1 \mathrm{gr}^0_{\mathrm{mot}} K(T) \to \pi_0 F^1 \xrightarrow{(i_1)_*} K_0(T) \xrightarrow{\mathrm{rk}} \mathbf{Z}$. This proves the first assertion of (iii) and defines $\widetilde{\lambda}$ without ambiguity.

It remains to calculate $(e_1)_*(\widetilde{\lambda})$. Since T is affine, integral, noetherian, regular, and one-dimensional, its coordinate ring is a Dedekind domain. Every invertible module over a Dedekind domain is generated by two elements. Choosing two generators of $\mathcal{L}$ gives a quotient $\mathcal{O}_T^{\oplus 2} \twoheadrightarrow \mathcal{L}$, hence a morphism $f\colon T \to \mathbf{P}^1_{k_0}$ such that $f^*\mathcal{O}_{\mathbf{P}^1_{k_0}}(1) \simeq \mathcal{L}$.

We now apply [Bou25, Constr. 4.3 and Rem. 4.4] with $X = \mathrm{Spec}(k_0)$ and $E = K$. If $\pi\colon \mathbf{P}^1_{k_0} \to \mathrm{Spec}(k_0)$ is the projection, the second summand of the filtered projective-bundle map is a morphism $u\colon \mathrm{Fil}^{\bullet-1}_{\mathrm{mot}} K(k_0) \to \mathrm{Fil}^{\bullet}_{\mathrm{mot}} K(\mathbf{P}^1_{k_0})$. At filtration degree one it gives $u_1\colon \mathrm{Fil}^0_{\mathrm{mot}} K(k_0) \to \mathrm{Fil}^1_{\mathrm{mot}} K(\mathbf{P}^1_{k_0})$. For K-theory, the line-bundle operator in Construction 4.3 is induced by the exact autoequivalence $- \otimes \mathcal{O}(-1)$. Hence the composite $\pi_0 \mathrm{Fil}^0_{\mathrm{mot}} K(k_0) \xrightarrow{(u_1)_*} \pi_0 \mathrm{Fil}^1_{\mathrm{mot}} K(\mathbf{P}^1_{k_0}) \to K_0(\mathbf{P}^1_{k_0})$ sends 1 to $1 - [\mathcal{O}(-1)]$.

On associated graded pieces, [Bou25, Rem. 4.4] identifies $\mathrm{gr}^1(u)$, after the shifts $\mathrm{gr}^j_{\mathrm{mot}} K \simeq \mathbf{Z}(j)_{\mathrm{mot}}[2j]$, with the second map in [Bou25, (4.2.1)]. By [Bou25, Def. 4.1 and Rem. 4.2], this map is

$$c_1^{\mathrm{mot}}(\mathcal{O}(1))\pi^*\colon \mathbf{Z}(0)_{\mathrm{mot}}(k_0) \to \mathbf{Z}(1)_{\mathrm{mot}}(\mathbf{P}^1_{k_0})[2].$$

Therefore, if $\eta := (u_1)_*(1) \in \pi_0 \mathrm{Fil}^1_{\mathrm{mot}} K(\mathbf{P}^1_{k_0})$, then its image under $\pi_0 \mathrm{Fil}^1_{\mathrm{mot}} K(\mathbf{P}^1_{k_0}) \to \pi_0 \mathrm{gr}^1_{\mathrm{mot}} K(\mathbf{P}^1_{k_0})$ is $c_1^{\mathrm{mot}}(\mathcal{O}(1))$. Functoriality of the motivic filtration gives $\eta_{\mathcal{L}} := f^*(\eta) \in \pi_0 F^1$. The two calculations above and

naturality of c_1^{mot} give $(i_1)_*(\eta_{\mathcal{L}}) = 1 - [\mathcal{L}^\vee]$ and $(e_1)_*(\eta_{\mathcal{L}}) = c_1^{\mathrm{mot}}(\mathcal{L})$. Let $b_{\mathcal{L}} := [\mathcal{L}] \cdot \eta_{\mathcal{L}} \in \pi_0 F^1$, where the product is the map $\pi_0 F^0 \otimes \pi_0 F^1 \to \pi_0 F^1$ induced by the multiplicative motivic filtration. Then $(i_1)_*(b_{\mathcal{L}}) = [\mathcal{L}](1-[\mathcal{L}^\vee]) = [\mathcal{L}]-1$, so injectivity of $(i_1)_*$ gives $b_{\mathcal{L}} = \widetilde{\lambda}$. The image of $[\mathcal{L}]$ in $\pi_0 \mathrm{gr}^0_{\mathrm{mot}} K(T) = H^0_{\mathrm{mot}}(T, \mathbf{Z}(0)) = \mathbf{Z}$ is its rank, namely 1. Multiplicativity on associated graded pieces therefore gives $(e_1)_*(\widetilde{\lambda}) = c_1^{\mathrm{mot}}(\mathcal{L})$. This proves (iii).

Finally, in the equicharacteristic situation at hand the Beilinson-Lichtenbaum map is a map of graded $\mathbf{E}_\infty$-algebras and is compatible with the first Chern class; see [EM23, Thm. 1.1 and Remark 4.42] and [BEM25, Thm. 7.16]. In weight one, $H^0(\mathrm{comp}_1)\colon \mu_3(T) \to H^0_{\text{Ét}}(T, \mu_3) = \mu_3(T)$ is the identity, while $H^2(\mathrm{comp}_1)$ sends the motivic first Chern class modulo 3 to the boundary class of $\mathcal{L}$ for $1 \to \mu_3 \to \mathbf{G}_m \xrightarrow{3} \mathbf{G}_m \to 1$. These are precisely the two identities in (iv), and multiplicativity gives $\mathrm{comp}_2(\zeta \smile (c_1^{\mathrm{mot}}(\mathcal{L}) \bmod 3)) = \zeta \smile \kappa(\mathcal{L})$. This proves (iv). □

Lemma 2.2. *Let T be a connected regular affine noetherian scheme of dimension one, ind-smooth over a field of characteristic zero. Suppose that $\Gamma(T, \mathcal{O}_T)$ contains a primitive cube root of unity ζ. Let $\mathcal{L}$ be a line bundle whose class is nonzero in $\mathrm{Pic}(T)/3$, and let $\beta \in K_2(T; \mathbf{Z}/3)$ satisfy $\partial_T(\beta) = [\zeta] \in K_1(T)[3]$. Then $\beta([\mathcal{L}] - 1) \neq 0$ in $K_2(T; \mathbf{Z}/3)$.*

Proof. Use the notation $F^q, F_3^q, i_q, i_{q,3}, e_q, e_{q,3}$ of Lemma 2.1. Define $\widetilde{\beta} := (i_{1,3})_*^{-1}(\beta) \in \pi_2 F_3^1$, and let $\widetilde{\lambda} \in \pi_0 F^1$ be the unique class such that $(i_1)_*(\widetilde{\lambda}) = [\mathcal{L}] - 1$. By parts (ii) and (iii) of that lemma, $(e_{1,3})_*(\widetilde{\beta}) = \zeta$ and $(e_1)_*(\widetilde{\lambda}) = c_1^{\mathrm{mot}}(\mathcal{L})$.

The module multiplication of part (i), at bidegree $(1,1)$, is a map $F_3^1 \wedge F^1 \to F_3^2$; it defines $\widetilde{\beta}\,\widetilde{\lambda} \in \pi_2 F_3^2$. Its image under $(i_{2,3})_*$ is $\beta([\mathcal{L}] - 1)$. The induced pairing on associated graded pieces is the motivic cup product, so

$$(e_{2,3})_*(\widetilde{\beta}\,\widetilde{\lambda}) = \zeta \smile \big(c_1^{\mathrm{mot}}(\mathcal{L}) \bmod 3\big) \in H^2_{\mathrm{mot}}(T, \mathbf{Z}/3(2)). \tag{5}$$

The long exact sequence of the weight-one coefficient triangle contains $\mathrm{Pic}(T) \xrightarrow{3} \mathrm{Pic}(T) \to H^2_{\mathrm{mot}}(T, \mathbf{Z}/3(1))$. Hence the second map induces an injection $\mathrm{Pic}(T)/3 \hookrightarrow H^2_{\mathrm{mot}}(T, \mathbf{Z}/3(1))$, and the hypothesis on $\mathcal{L}$ gives $c_1^{\mathrm{mot}}(\mathcal{L}) \bmod 3 \neq 0$.

Since 3 is invertible on T, Beilinson-Lichtenbaum gives an isomorphism $H^2_{\mathrm{mot}}(T, \mathbf{Z}/3(2)) \xrightarrow{\sim} H^2_{\text{Ét}}(T, \mu_3^{\otimes 2})$; see [Bou24, Cor. 5.6]. By part (iv) of Lemma 2.1, the image of (5) under this isomorphism is $\zeta \smile \kappa(\mathcal{L})$.

The Kummer sequence gives the exact segment $\mathrm{Pic}(T) \xrightarrow{3} \mathrm{Pic}(T) \xrightarrow{\kappa} H^2_{\text{Ét}}(T, \mu_3)$, so κ induces an injection $\mathrm{Pic}(T)/3 \hookrightarrow H^2_{\text{Ét}}(T, \mu_3)$. In particular, $\kappa(\mathcal{L}) \neq 0$. Since ζ is primitive, the map of étale sheaves $\mu_3 \to \mu_3^{\otimes 2}$, $\quad s \mapsto \zeta \otimes s$ is an isomorphism. The induced map on $H^2_{\text{Ét}}(T, -)$ sends $\kappa(\mathcal{L})$ to $\zeta \smile \kappa(\mathcal{L})$. Consequently $\zeta \smile \kappa(\mathcal{L}) \neq 0$, and (5) implies $(e_{2,3})_*(\widetilde{\beta}\,\widetilde{\lambda}) \neq 0$. Thus $\widetilde{\beta}\,\widetilde{\lambda} \neq 0$ in $\pi_2 F_3^2$. It remains to verify that the image of $\widetilde{\beta}\,\widetilde{\lambda}$ in $K_2(T; \mathbf{Z}/3)$ is nonzero. The weight-one identification $\mathbf{Z}(1)_{\mathrm{mot}}(T) \simeq R\Gamma_{\mathrm{Zar}}(T, \mathbf{G}_m)[-1]$ and its coefficient triangle give $H^{-1}_{\mathrm{mot}}(T, \mathbf{Z}/3(1)) = 0$. By (4), this is the equality $\pi_3 \mathrm{gr}^1_{\mathrm{mot}} K(T; \mathbf{Z}/3) = 0$. The fibre sequence $F_3^2 \to F_3^1 \to \mathrm{gr}^1_{\mathrm{mot}} K(T; \mathbf{Z}/3)$ therefore shows that $\pi_2 F_3^2 \to \pi_2 F_3^1$ is injective. Part (ii) of Lemma 2.1 identifies $\pi_2 F_3^1$ with $K_2(T; \mathbf{Z}/3)$, and the image of $\widetilde{\beta}\,\widetilde{\lambda}$ under this composite injection is $\beta([\mathcal{L}] - 1)$. Hence $\beta([\mathcal{L}] - 1) \neq 0$, as claimed. Notice that the argument uses neither completeness nor separatedness of the motivic filtration. □

3. The cyclotomic A_2 surface

Choose a primitive cube root of unity ζ, put $\pi := \zeta - 1$, and set $O := V[\zeta] = V[\pi]$. The element π satisfies $\pi^2 + 3\pi + 3 = 0$. Since 3 is a uniformizer of V, this equation is Eisenstein. Hence O is a discrete valuation ring, finite free and totally ramified of degree two over V, with maximal ideal (π). Its nontrivial automorphism is denoted by σ. Set

$$t := \zeta^2 - \zeta, \qquad u := x - t, \qquad v := x + t.$$

Since $t^2 = -3$, base change of (2) gives

$$B := A \otimes_V O \simeq O[\![x, y]\!]/(uv - y^3). \tag{6}$$

Let $P := (u, y)$ and $Q := (v, y)$ be the height-one prime ideals of B. Put $R := B[1/3]$ and $T := \mathrm{Spec}(R)$.

Our divisor-class convention is that $[P] \in \mathrm{Cl}(B)$ denotes the class of the prime Weil divisor $V(P)$. By contrast, $P|_T$ denotes the restriction to T of the rank-one ideal sheaf P, which corresponds to $\mathcal{O}_T(-V(P)|_T)$.

Thus the two conventions differ by the usual sign when divisor classes are identified with rank-one reflexive sheaves.

Proposition 3.1. *The ring B is a complete normal local domain, finite free of rank two over A. Its divisor class group is cyclic of order three, generated by $[P]$, $\operatorname{div}(u) = 3P$, and $\operatorname{div}(y) = P + Q$. If $\mathcal{L} := P|_T$, then T is integral, regular and one-dimensional,* $\operatorname{Pic}(T) \simeq \mathbf{Z}/3$*, the class of $\mathcal{L}$ is a generator, and $Q|_T \simeq \mathcal{L}^{-1}$.*

Proof. Let $\mathfrak{n} := (3, x, y)$ in $V[\![x, y]\!]$. The ring $V[\![x, y]\!]$ is a three-dimensional complete regular local ring, and since 3 is a uniformizer of V, one has $3 \notin \mathfrak{n}^2$, and $3 + x^2 - y^3 \equiv 3 \pmod{\mathfrak{n}^2}$. Thus the defining equation is part of a minimal system of generators of $\mathfrak{n}$, so $A = V[\![x, y]\!]/(3 + x^2 - y^3)$ is a two-dimensional regular local domain [The26, Tags 032C, 00NQ and 00NP]. As a quotient of the complete noetherian local ring $V[\![x, y]\!]$, the ring A is complete. Its maximal ideal is (x, y), and $3 = y^3 - x^2 \in (x, y)^2$, which is precisely the asserted ramification. Since O is free of rank two over V, the identity $B = A \otimes_V O$ makes B finite free of rank two over A. The presentation (6) makes it a complete local ring [The26, Tag 0325]. For this uniformizer one has $t = \zeta\pi$ and $\pi^2 = -3\zeta$. Since A is a domain of characteristic zero, it is V-torsion-free and hence flat over the discrete valuation ring V [The26, Tag 0539]. Consequently $B = A \otimes_V O$ is O-flat, so π is a nonzerodivisor. Reduction modulo π gives

$$B/\pi B \simeq k[\![x, y]\!]/(x^2 - y^3) \simeq k[\![s^2, s^3]\!], \qquad x \longmapsto s^3, \quad y \longmapsto s^2,$$

which is a domain.

Krull's intersection theorem gives $\bigcap_n \mathfrak{m}_B^n = 0$, and $\pi^n B \subseteq \mathfrak{m}_B^n$, so every nonzero $b \in B$ lies in $\pi^r B$ for a largest $r \geq 0$. If $b = \pi^r b_0$ and $c = \pi^s c_0$ are nonzero with $b_0, c_0 \notin \pi B$ and $bc = 0$, then cancelling the nonzerodivisor π^{r+s} gives $b_0 c_0 = 0$, whose reduction modulo π contradicts the display above. Hence B is a domain.

Write $B \simeq O[\![u, y]\!]/(u^2 + 2tu - y^3)$. The ambient ring is regular local of dimension three and the displayed equation is a nonzerodivisor. Hence B is a two-dimensional Cohen-Macaulay hypersurface, and in particular satisfies S_2 [The26, Tags 00NQ, 02JN and 0342].

Moreover $B[1/\pi] \simeq A[1/3] \otimes_{V[1/3]} O[1/3]$. The ring $A[1/3]$ is regular, while $O[1/3]/V[1/3]$ is finite étale; regularity is preserved by localization and étale base change [The26, Tags 0AFS and 0AH0]. Hence $B[1/\pi]$ is regular. It is one-dimensional: every prime not containing π has height at most one, since the unique height-two prime $\mathfrak{m}_B$ contains π, while P survives after inverting π, because $B/P \simeq O$ and $\pi \neq 0$ in O.

Since $B/\pi B$ is a domain, (π) is prime; it has height one, and $B_{(\pi)}$ is one-dimensional with maximal ideal generated by π, so it is a discrete valuation ring. Therefore B satisfies R_1 and S_2, and Serre's criterion shows that it is normal [The26, Tag 031O].

We next compute $\operatorname{Cl}(B)$. Since $B/(u) \simeq O[\![y]\!]/(y^3)$, the only height-one prime containing u is P. In B_P, the elements t and $v = u + 2t$ are units, $PB_P = (y)$, and $u = y^3/v$. Thus $\operatorname{div}(u) = 3P$. Similarly, $B/(y) \simeq O[\![u]\!]/(u(u + 2t))$, and localization at either of its two minimal primes shows that $\operatorname{div}(y) = P + Q$. Hence $3[P] = 0$ and $[P] + [Q] = 0$. Moreover, $[P] \neq 0$. Indeed, if P were principal, its image modulo π would be principal. Since $u = x - t$ and $t = \zeta\pi$, this image is the maximal ideal $(\bar{x}, \bar{y})$ of $k[\![x, y]\!]/(x^2 - y^3)$, whereas $(\bar{x}, \bar{y})/(\bar{x}, \bar{y})^2$ has k-dimension two. Thus $[P]$ has exact order three.

It remains to prove that P generates the whole class group.[2] Let $\rho\colon \widetilde{X} \to \operatorname{Spec}(B)$ be the blow-up of $\mathfrak{m}_B = (\pi, x, y)$; it is projective, hence proper [The26, Tags 02NS/01WC]. It is an isomorphism over $\operatorname{Spec}(B) \setminus \{\mathfrak{m}_B\}$, which is regular because B is two-dimensional and satisfies R_1; so only the points of the exceptional divisor have to be examined. Since $\operatorname{Spec}(B)$ is a hypersurface in $\operatorname{Spec}(O[\![x, y]\!])$ and the strict transform of a closed subscheme is its blow-up in the induced centre [The26, Tag 080E], $\widetilde{X}$ is the strict transform of $\operatorname{Spec}(B)$ in the blow-up of $\operatorname{Spec}(O[\![x, y]\!])$ at its maximal ideal, which is regular of dimension three; in each chart below $\widetilde{X}$ is therefore cut out by one equation in a three-dimensional regular local ring. Since $\pi^2 = -3\zeta$, the equation of B is $x^2 - \zeta^2\pi^2 - y^3 = 0$, whose initial form is $X^2 - \Pi^2$, since $\zeta \equiv 1 \bmod \pi$. This form is a nonzerodivisor in $k[X, \Pi, Y]$, so it generates the initial ideal and $\operatorname{gr}_{\mathfrak{m}_B}(B) \simeq k[X, \Pi, Y]/(X^2 - \Pi^2)$. The exceptional divisor is therefore the reduced union $E = E_1 \cup E_2$ of the lines $X = \Pi$ and $X = -\Pi$ in $\mathbf{P}^2_k$, meeting transversally at $e = [0 : 0 : 1]$.

On the Π-chart the strict transform has equation $(x_1 - \zeta)(x_1 + \zeta) = \pi y_1^3$. Along the exceptional divisor exactly one of the two factors lies in the maximal ideal, the other being a unit because 2ζ is a unit; hence the equation is part of a regular system of parameters and the local ring is regular. This chart contains $E \setminus \{e\}$.

[2] This is the direct A_2 instance of Lipman's discriminant-group description of class groups of rational surface singularities; compare [Lip69, Prop. 17.1].

At e we pass to the Y-chart, where $x = yx_3$ and $\pi = y\pi_3$; eliminating y by the equation $y = x_3^2 - \zeta^2\pi_3^2$ of the strict transform, the completed local ring there is $O[\![x_3, \pi_3]\!]/(\pi - (x_3^2 - \zeta^2\pi_3^2)\pi_3)$, which is regular because its defining equation is congruent to π modulo the square of the maximal ideal. Hence $\widetilde{X}$ is regular [The26, Tag 07NY].

In that chart, E is cut out by $y = (x_3 - \zeta\pi_3)(x_3 + \zeta\pi_3)$, and $(x_3 - \zeta\pi_3, x_3 + \zeta\pi_3) = (x_3, \pi_3)$ is the maximal ideal; so E_1 and E_2 meet transversally at e. On the Π-chart they are disjoint, because 2ζ is a unit. Hence the scheme-theoretic intersection $E_1 \cap E_2$ is the reduced point e, whose residue field is k.

Each E_i is a line in $\mathbf{P}^2_k$, hence a proper k-scheme of dimension one, so every invertible $\mathcal{O}_{E_i}$-module $\mathcal{N}$ has a degree $\deg_{E_i}(\mathcal{N}) = \chi(E_i, \mathcal{N}) - \chi(E_i, \mathcal{O}_{E_i})$ [The26, Tag 0AYR], and $\deg_{E_i}$ is a homomorphism $\mathrm{Pic}(E_i) \to \mathbf{Z}$ [The26, Tag 0AYX]. For a Cartier divisor D on $\widetilde{X}$ put $D \cdot E_i := \deg_{E_i}\big(\mathcal{O}_{\widetilde{X}}(D)|_{E_i}\big)$, which is therefore additive in D.

By construction of the blow-up, $\mathfrak{m}_B\mathcal{O}_{\widetilde{X}}$ is the ideal of the effective Cartier divisor E, and $\mathcal{O}_{\widetilde{X}}(-E) = \mathcal{O}_{\widetilde{X}}(1)$ [The26, Tag 02OS]. Under the closed immersion $E = \mathrm{Proj}(\mathrm{gr}_{\mathfrak{m}_B}(B)) \subset \mathbf{P}^2_k$ one has $\mathcal{O}_{\widetilde{X}}(1)|_E = \mathcal{O}_{\mathbf{P}^2}(1)|_E$, whose restriction to the line $E_i \simeq \mathbf{P}^1_k$ is $\mathcal{O}_{\mathbf{P}^1}(1)$, of degree one. Hence $E \cdot E_i = -1$ for $i = 1, 2$.

Moreover $E_2 \not\subset E_1$, so the pullback of E_1 along $E_2 \hookrightarrow \widetilde{X}$ is an effective Cartier divisor on E_2 and $\mathcal{O}_{\widetilde{X}}(E_1)|_{E_2} = \mathcal{O}_{E_2}(E_1 \cap E_2)$. By the chart computations above, $E_1 \cap E_2$ is the reduced point e with residue field k, so $E_1 \cdot E_2 = \dim_k \Gamma(E_1 \cap E_2, \mathcal{O}) = 1$ [The26, Tag 0AYY]; exchanging the roles of E_1 and E_2 gives $E_2 \cdot E_1 = 1$. Finally E is reduced with irreducible components E_1 and E_2, and $\widetilde{X}$ is regular, so $E = E_1 + E_2$ as Cartier divisors. Additivity in the first variable therefore yields $E_1^2 = E \cdot E_1 - E_2 \cdot E_1 = -2$ and $E_2^2 = E \cdot E_2 - E_1 \cdot E_2 = -2$.

For $n \geq 1$ let $E^{(n)} \subset \widetilde{X}$ be the closed subscheme defined by $\mathcal{O}_{\widetilde{X}}(-nE)$, so that $E^{(1)} = E$ and $E^{(n)} = \widetilde{X} \times_{\mathrm{Spec}(B)} \mathrm{Spec}(B/\mathfrak{m}_B^n)$. Let $\nu\colon E_1 \sqcup E_2 \to E$ be the normalization. The sequence

$$1 \to \mathcal{O}_E^\times \to \nu_*\mathcal{O}^\times \to k^\times(e) \to 1$$

is exact and $H^0(\nu_*\mathcal{O}^\times) = k^\times \times k^\times \to k^\times(e)$ is surjective, so multidegree identifies $\mathrm{Pic}(E)$ with $\mathbf{Z}^2$.

Tensoring the additive normalization sequence $0 \to \mathcal{O}_E \to \nu_*\mathcal{O} \to k(e) \to 0$ with $\mathcal{O}_E(-nE)$ gives a sequence whose restrictions to the two components are $\mathcal{O}_{\mathbf{P}^1}(n)$. Since evaluation at e is surjective on global sections and $H^1(\mathbf{P}^1, \mathcal{O}(n)) = 0$ for $n \geq 0$, it follows that $H^1(E, \mathcal{O}_E(-nE)) = 0$. Also $H^2(E, \mathcal{O}_E(-nE)) = 0$, since E is a curve. For $n \geq 1$, the morphism $E^{(n)} \to E^{(n+1)}$ is a square-zero thickening with ideal $\mathcal{I}_n := \mathcal{O}_{\widetilde{X}}(-nE)/\mathcal{O}_{\widetilde{X}}(-(n+1)E) \simeq \mathcal{O}_E(-nE)$. The exact sequence of Zariski sheaves

$$1 \longrightarrow 1 + \mathcal{I}_n \longrightarrow \mathcal{O}^\times_{E^{(n+1)}} \longrightarrow \mathcal{O}^\times_{E^{(n)}} \longrightarrow 1, \qquad 1 + \mathcal{I}_n \simeq \mathcal{I}_n,$$

gives an exact segment $H^1(E, \mathcal{I}_n) \to \mathrm{Pic}(E^{(n+1)}) \to \mathrm{Pic}(E^{(n)}) \to H^2(E, \mathcal{I}_n)$ [The26, Tag 0C6R]. The two outer groups vanish by the previous paragraph, so $\mathrm{Pic}(E^{(n+1)}) \simeq \mathrm{Pic}(E^{(n)})$. The same vanishing of $H^1(E, \mathcal{I}_n)$ makes the restriction maps on global units surjective, so isomorphisms between compatible systems of line bundles can be chosen compatibly.

Let $(\mathcal{L}_n)_{n \geq 1}$ be a compatible system with $\mathcal{L}_n \in \mathrm{Pic}(E^{(n)})$. Since B is noetherian and $\mathfrak{m}_B$-adically complete and $\rho\colon \widetilde{X} \to \mathrm{Spec}(B)$ is proper, Grothendieck existence identifies the completion functor

$$\mathrm{Coh}(\widetilde{X}) \xrightarrow{\sim} \mathrm{Coh}\big(\widetilde{X}, \mathfrak{m}_B\mathcal{O}_{\widetilde{X}}\big), \qquad \mathcal{F} \longmapsto \big(\mathcal{F}/\mathfrak{m}_B^n\mathcal{F}\big)_{n \geq 1};$$

see [The26, Tag 088C]. Here $\mathfrak{m}_B^n\mathcal{O}_{\widetilde{X}} = \mathcal{O}_{\widetilde{X}}(-nE)$, so the n-th quotient is a coherent sheaf on $E^{(n)}$. Consequently the system $(\mathcal{L}_n)_n$ is the completion of a coherent $\mathcal{O}_{\widetilde{X}}$-module $\mathcal{F}$, unique up to isomorphism.

For any $z \in E$, $\mathcal{F}_z \simeq \mathcal{O}_{\widetilde{X},z}$ is locally free of rank one. Since $\widetilde{X}$ is noetherian, $\mathcal{F}$ is finitely presented, and its locally-free-of-rank-one locus is open [The26, Tag 0GWM]. Let $Z \subset \widetilde{X}$ be its closed complement. Then $Z \cap E = \varnothing$.

The composite $Z \longrightarrow \widetilde{X} \xrightarrow{\rho} \mathrm{Spec}(B)$ is proper, so $\rho(Z)$ is closed [The26, Tag 01W6]. If Z were nonempty, the nonempty closed subset $\rho(Z)$ of the local scheme $\mathrm{Spec}(B)$ would contain its closed point. Its inverse image would then meet the closed fibre E, contradicting $Z \cap E = \varnothing$. Therefore $Z = \varnothing$, and $\mathcal{F}$ is invertible. Essential surjectivity and full faithfulness in Grothendieck existence therefore give

$$\mathrm{Pic}(\widetilde{X}) \xrightarrow{\sim} \varprojlim_n \mathrm{Pic}(E^{(n)}) \xrightarrow{\sim} \mathrm{Pic}(E) \xrightarrow{\sim} \mathbf{Z}^2.$$

Put $\widetilde{X}^\circ := \widetilde{X} \setminus E$. By [The26, Tag 02OS] the morphism ρ restricts to an isomorphism $\widetilde{X}^\circ \simeq \mathrm{Spec}(B) \setminus \{\mathfrak{m}_B\}$; as noted at the beginning of this argument, these schemes are regular. The scheme $\widetilde{X}$ is integral [The26,

Tag 02ND], noetherian and regular, and so is its open subscheme $\widetilde{X}^\circ$; their local rings are unique factorization domains [The26, Tag 0AG0], whence $\mathrm{Pic} \xrightarrow{\sim} \mathrm{Cl}$ for both [The26, Tag 0BE9].

We use the presentation of Cl by Weil divisors modulo principal divisors [The26, Tags 0BE2, 0BE3 and 0BE4]. Taking closures identifies the prime divisors of $\widetilde{X}^\circ$ with the prime divisors of $\widetilde{X}$ meeting $\widetilde{X}^\circ$, so restriction $\mathrm{Div}(\widetilde{X}) \to \mathrm{Div}(\widetilde{X}^\circ)$ is surjective with kernel free on the prime divisors contained in E, namely E_1 and E_2. As $\widetilde{X}^\circ$ is dense in $\widetilde{X}$ and orders of vanishing are computed at generic points, this restriction carries $\mathrm{div}(f)$ to $\mathrm{div}(f)$ for every f in the common function field. Hence $\mathbf{Z}E_1 \oplus \mathbf{Z}E_2 \to \mathrm{Pic}(\widetilde{X}) \to \mathrm{Pic}(\widetilde{X}^\circ) \to 0$ is exact, the first map sending E_j to $\mathcal{O}_{\widetilde{X}}(E_j)$.

The identification $\mathrm{Pic}(\widetilde{X}) \simeq \mathbf{Z}^2$ obtained above is $\mathcal{M} \mapsto \big(\deg_{E_1}(\mathcal{M}|_{E_1}), \deg_{E_2}(\mathcal{M}|_{E_2})\big)$, so by the definition of the intersection numbers it sends $\mathcal{O}_{\widetilde{X}}(E_j)$ to $(E_j \cdot E_1, E_j \cdot E_2)$. The first map of the exact sequence above is therefore given by the matrix $M := \left(\begin{smallmatrix} E_1\cdot E_1 & E_2\cdot E_1 \\ E_1\cdot E_2 & E_2\cdot E_2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$, whose j-th column is the image of E_j.

Write ϵ_1, ϵ_2 for the images in $\mathrm{coker}(M)$ of the standard basis of $\mathbf{Z}^2$. The two columns of M impose the relations $-2\epsilon_1 + \epsilon_2 = 0$ and $\epsilon_1 - 2\epsilon_2 = 0$; the first gives $\epsilon_2 = 2\epsilon_1$, and substituting it into the second gives $3\epsilon_1 = 0$. Thus $\mathrm{coker}(M)$ is cyclic of order dividing three, generated by ϵ_1. It is exactly of order three, since the surjection $\mathbf{Z}^2 \to \mathbf{Z}/3$, $(a, b) \mapsto a + 2b$, kills both columns of M. Consequently $\mathrm{Pic}(\widetilde{X}^\circ) \simeq \mathrm{coker}(M) \simeq \mathbf{Z}/3$.

Finally, no prime divisor of $\mathrm{Spec}(B)$ is contained in $\{\mathfrak{m}_B\}$, so the same restriction argument, applied to $\mathrm{Spec}(B) \setminus \{\mathfrak{m}_B\} \subset \mathrm{Spec}(B)$, gives $\mathrm{Cl}(B) \xrightarrow{\sim} \mathrm{Cl}(\widetilde{X}^\circ) = \mathrm{Pic}(\widetilde{X}^\circ) \simeq \mathbf{Z}/3$. Since $[P]$ has exact order three, it is a generator.

We have $B[1/3] = B[1/\pi]$. Restriction of Weil divisors to $D(\pi)$ is surjective, and its kernel is generated by the height-one primes containing π. The only such prime is (π), whose class is zero because it is the principal divisor of π. The presentation of the divisor class group by Weil divisors and principal divisors therefore gives $\mathrm{Cl}(B) \xrightarrow{\sim} \mathrm{Cl}(B[1/3])$ [The26, Tags 0BE2, 0BE3 and 0BE4]. The ring $B[1/3]$ is regular, so $\mathrm{Pic}(T) \xrightarrow{\sim} \mathrm{Cl}(B[1/3])$ [The26, Tags 0AG0 and 0BE9]. Hence $\mathrm{Pic}(T) \simeq \mathbf{Z}/3$ and $\mathcal{L} = P|_T$ is a generator. Since $\mathrm{div}(y) = P + Q$, we have $Q|_T \simeq (P|_T)^{-1} = \mathcal{L}^{-1}$. □

4. Construction of the generic-kernel class

If R is noetherian, write $G(R)$ for the K-theory of finitely generated R-modules; if R is regular, $K(R) \to G(R)$ is an equivalence. A finite ring map $f\colon R \to S$ gives a pushforward $f_*\colon G(S) \longrightarrow G(R)$ by restriction of scalars, while a flat map gives a pullback by base change.

Lemma 4.1. *The element y is prime in A, and there is an isomorphism of V-algebras*

$$A/yA \xrightarrow{\sim} O, \qquad x \longmapsto t.$$

Proof. Since $t = -1 - 2\zeta$ and $2 \in V^\times$, one has $O = V[\zeta] = V[t]$; moreover $t^2 = -3$.

Since $t = \zeta\pi$ lies in the maximal ideal of the complete local ring O, substitution $x \mapsto t$ defines a continuous V-algebra map $V[\![x]\!] \to O$. As $t^2 = -3$, it induces a surjection $V[\![x]\!]/(x^2 + 3) \to O$, the surjectivity following from $O = V[t]$. By Weierstrass division the source is free over V with basis $1, x$, while $1, t$ is a V-basis of O. The map is therefore an isomorphism.

Hence $A/yA \simeq O$ is a domain, so yA is prime. □

Since O is commutative local, determinant identifies $K_1(O)$ with $O^\times$ [Wei13, Lem. III.1.4], and the unit ζ has order three. Exactness of the coefficient sequence for O therefore supplies a class $\beta_0 \in K_2(O; \mathbf{Z}/3)$ with $\partial_O(\beta_0) = [\zeta]$. Put

$$\beta := \frac{1}{2}\big(\beta_0 - \sigma^*\beta_0\big) \in K_2(O; \mathbf{Z}/3). \tag{7}$$

Because $\sigma(\zeta) = \zeta^2$ and $\frac{1}{2} = 2$ in $\mathbf{Z}/3$, one has $\partial_O(\beta) = 2\big([\zeta] - 2[\zeta]\big) = -2[\zeta] = [\zeta]$ and $\sigma^*\beta = -\beta$.

Let $i\colon \mathrm{Spec}(A/yA) \to \mathrm{Spec}(A)$ be the closed immersion and $i_*\colon G(A/yA; \mathbf{Z}/3) \to G(A; \mathbf{Z}/3)$ the induced pushforward. Using Lemma 4.1 to read β as a class on $\mathrm{Spec}(A/yA)$, set

$$a := i_*(\beta) \in G_2(A; \mathbf{Z}/3) = K_2(A; \mathbf{Z}/3). \tag{8}$$

Lemma 4.2. *Write $K_O := \mathrm{Frac}(O)$. The ring $R = B[1/3]$ is flat over A, and $R/yR \xrightarrow{\sim} K_O \times K_O$, the two factors being R/PR and R/QR. Under Lemma 4.1 and the identifications $B/P \simeq O \simeq B/Q$ induced by $O \subset B$, the map $A/yA \to R/yR$ is $\lambda \mapsto \big(\lambda, \sigma(\lambda)\big)$.*

Proof. By Proposition 3.1 the ring B is finite free over A, hence flat, and R is a localization of B; so R is flat over A. Setting $y = 0$ in (6) gives $B/yB \simeq O[\![x]\!]/\big((x-t)(x+t)\big)$. Since $t^2 = -3$, the element $2t$ is a unit after inverting 3, so $(x-t)$ and $(x+t)$ are comaximal in $(B/yB)[1/3]$ and the Chinese remainder theorem gives

$$R/yR \simeq \big(O[\![x]\!]/(x-t)\big)[1/3] \times \big(O[\![x]\!]/(x+t)\big)[1/3] \simeq K_O \times K_O,$$

the two projections sending x to t and to $-t$. As $u = x - t$ and $v = x + t$, the first factor is R/PR and the second is R/QR. Finally A/yA is generated over V by the class of x, which corresponds to $t \in O$ under Lemma 4.1, and $\sigma(t) = -t$; so the map $A/yA \to R/yR$ is $\lambda \mapsto (\lambda, \sigma\lambda)$. □

Proposition 4.3. *The class a is nonzero, its image in $K_2(F; \mathbf{Z}/3)$ is zero, and $\partial_A(a) = 0$.*

Proof. Since y is a nonzerodivisor, Quillen's localization sequence

$$G(A/yA; \mathbf{Z}/3) \xrightarrow{i_*} G(A; \mathbf{Z}/3) \longrightarrow G(A[1/y]; \mathbf{Z}/3)$$

is exact [Qui73, §7], so a dies in $G_2(A[1/y]; \mathbf{Z}/3)$. As F is a localization of $A[1/y]$, and as A and F are regular so that the resolution theorem identifies the G-groups occurring here with their K-groups [Wei13, Thm. V.3.3], the image of a in $K_2(F; \mathbf{Z}/3)$ is zero.

We prove that $a \neq 0$. Let $g\colon A \to R$ be as in Lemma 4.2. For a finite A/yA-module M one has $M \otimes_A R = M \otimes_{A/yA} (R/yR)$, and both functors are exact because R is flat over A; with Lemma 4.2 this gives

$$g^* i_* = i_{P*}\varphi_P^* + i_{Q*}\varphi_Q^*,$$

where i_P, i_Q are the closed immersions of the two points of T cut out by P and Q, while $\varphi_P\colon O \hookrightarrow K_O$ is the standard inclusion and $\varphi_Q := \varphi_P \circ \sigma$; these are the two components of $A/yA \simeq O \to K_O \times K_O$.

Writing $\beta|_T$ and $\beta|_{K_O}$ for the images of β along $O \subset B \to R$ and $O \subset K_O$, and using $\sigma^*\beta = -\beta$, we obtain $g^*(a) = i_{P*}\big(\beta|_{K_O}\big) - i_{Q*}\big(\beta|_{K_O}\big)$. Both identifications $B/P \simeq O \simeq B/Q$ restrict to the identity on O, so $i_P^*(\beta|_T) = i_Q^*(\beta|_T) = \beta|_{K_O}$, and the projection formula [Wei13, Thm. V.3.12] gives $g^*(a) = \beta|_T \cdot \big([R/PR] - [R/QR]\big)$.

Put $\mathcal{L} := P|_T$. The exact sequences $0 \to \mathcal{L} \to R \to R/PR \to 0$ and $0 \to Q|_T \to R \to R/QR \to 0$ give $[R/PR] - [R/QR] = [Q|_T] - [\mathcal{L}]$ in $K_0(T)$. For a connected one-dimensional regular noetherian ring, rank and determinant induce an isomorphism $K_0(T) \simeq \mathbf{Z} \oplus \mathrm{Pic}(T)$ [Wei13, Cor. II.2.6.3], under which this difference corresponds to $(0, Q|_T \otimes \mathcal{L}^{-1})$. By Proposition 3.1, $Q|_T \simeq \mathcal{L}^{-1}$ and $\mathcal{L}^{\otimes 3} \simeq \mathcal{O}_T$, so $Q|_T \otimes \mathcal{L}^{-1} \simeq \mathcal{L}^{-2} \simeq \mathcal{L}$ and therefore $g^*(a) = \beta|_T\big([\mathcal{L}] - 1\big)$.

We check the hypotheses of Lemma 2.2 for T. By Proposition 3.1 it is integral, regular, noetherian and one-dimensional, and the class of $\mathcal{L}$ generates $\mathrm{Pic}(T) \simeq \mathbf{Z}/3$, hence is nonzero in $\mathrm{Pic}(T)/3$. The ring $\Gamma(T, \mathcal{O}_T)$ contains ζ, and 3 is invertible on T. If $K_O = \mathrm{Frac}(O)$, then R is a regular noetherian K_O-algebra; the field K_O is perfect because it has characteristic zero, so R is geometrically regular over K_O [The26, Tags 05DU and 038V], and Popescu's theorem writes R as a filtered colimit of smooth K_O-algebras [The26, Tag 07GC]. Finally $\partial_T(\beta|_T) = [\zeta]$ by naturality of the coefficient boundary. Lemma 2.2 therefore gives $g^*(a) \neq 0$, whence $a \neq 0$.

It remains to compute the Bockstein. Exactness of (1) shows that $\partial_A(a) \in K_1(A)[3]$, and determinant identifies $K_1(A)$ with $A^\times$ [Wei13, Lem. III.1.4]. Now $B \otimes_A F = F \otimes_V O = F[\zeta]$ is a domain, being a localization of B, and has F-dimension two because B is finite free of rank two over A (Proposition 3.1). A finite-dimensional domain over a field is a field, so $F[\zeta]$ is quadratic over F and $\zeta \notin F$. Also $\zeta^2 \notin F$, since otherwise $\zeta = (\zeta^2)^2$ would belong to F. Thus F, and hence A, contains no nontrivial cube root of unity. Therefore $K_1(A)[3] = K_1(F)[3] = 0$. The coefficient sequences identify $K_2(A; \mathbf{Z}/3) \simeq K_2(A)/3$ and $K_2(F; \mathbf{Z}/3) \simeq K_2(F)/3$; in particular, $\partial_A(a) = 0$.

□

Proof of Theorem 1.1. The first paragraph of the proof of Proposition 3.1 shows that A is a complete two-dimensional ramified regular local domain. The required class is the class a constructed in Proposition 4.3. □

5. Integral divisibility and possible odd-prime variants

The counterexample is represented by an integral class that becomes divisible by three over the fraction field.

Corollary 5.1. *There is $c \in K_2(A) \setminus 3K_2(A)$ lifting a and an element $d \in K_2(F)$ such that $c|_F = 3d$, but d is not in the image of $K_2(A)$.*

Proof. Since $\partial_A(a) = 0$, exactness of (1) gives an integral lift c. Since $a \neq 0$, necessarily $c \notin 3K_2(A)$. Since $a|_F = 0$, the same sequence over F gives $c|_F = 3d$ for some $d \in K_2(F)$. Van der Kallen proved that $K_2(A) \to K_2(F)$ is injective [vdK76, main theorem]. If $d = e|_F$ for some $e \in K_2(A)$, then $c - 3e$ would vanish over F and hence vanish in $K_2(A)$; taking the image in $K_2(A; \mathbf{Z}/3)$ would give $a = 0$, a contradiction. □

We record a proposed odd-prime extension in degree $p - 1$; we do not prove it here.

Remark 5.2. The same method is expected to yield the following variants.

Let p be odd, and $V = W(\overline{\mathbf{F}}_p)$. Let $\chi\colon \mathrm{Gal}(V[\zeta_p]/V) \to \mathbf{F}_p^\times$ be the cyclotomic character, and put $s = (p-1)/2$. Choose a uniformizer $t \in V[\zeta_p]^{\ker(\chi^s)}$ such that $\sigma(t) = \chi^s(\sigma)t$ for any $\sigma \in \mathrm{Gal}(V[\zeta_p]/V)$, and set $d = t^2 \in V$. Then $A_p := V[\![x, y]\!]/(x^2 - d - y^p)$ is a complete two-dimensional ramified regular local domain. The expected conclusion is that there exists

$$0 \neq a_p \in \ker\big(K_{p-1}(A_p; \mathbf{Z}/p) \longrightarrow K_{p-1}(\mathrm{Frac}(A_p); \mathbf{Z}/p)\big).$$

Remark 5.3. For $p = 3$, elementary computations using symbols could be used to bypass Section 2.[3]

We have kept the motivic argument: it exhibits the class as a weight-two phenomenon, and it is the one that should generalize to the higher-degree variants in K_{p-1} described in Remark 5.2 for odd p.

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[3] Put $g := -(1+x)/2$ and $s := \{g, y\} \in K_2(A[1/y])$. Since $g \bmod y = \zeta$, localization lifts $3s$ to $c \in K_2(A)$. If $c = 3e$, then $(s-e)|_L \in K_2(L)[3]$, where $L = \mathrm{Frac}(B)$, equals $\{\zeta, h\}$ for some $h \in L^\times$ by [Wei13, Thm. III.6.8]. As $e|_L$ is unramified on T, tame symbols give $\mathrm{div}_T(h) \equiv P - Q \pmod 3$, contradicting Proposition 3.1. Thus c mod 3 is a nonzero generic-kernel class.